MIL-E-17777D
27 February 1984
SUPERSEDING
MIL-E-17777C
9 August 1974
(See 6.4)

MILITARY SPECIFICATION

ELECTRODES, CUTTING AND WELDING, CARBON-GRAPHITE, UNCOATED AND COPPER-COATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers round section uncoated (plain) carbon-graphite electrodes for arc-welding, round, half-round and rectangular section copper-coated carbon-graphite electrodes for air carbon-arc cutting, and round section jointed copper-coated carbon-graphite electrodes for air carbon-arc cutting.

1.2 Classification. The electrodes shall be furnished in the types and sizes shown in table I, as specified (see 6.2.1).

TABLE I. Types and sizes.

Type	Description	Nominal size (inches)		
		Diameter	Cross-section	Length
MIL-C	Round carbon-graphite (plain-uncoated)	1/8, 5/32, 3/16, 1/4 5/16 3/8 1/2, 5/8, 3/4, 1	Circular	12 14 15 17
MIL-CuC-A	Round copper-coated carbon-graphite	1/8, 5/32, 3/16, 1/4 5/16 3/8 1/2, 5/8, 3/4, 1	Circular	12 14 15 17

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TABLE I. Types and sizes. - Continued

Type	Description	Nominal size (inches)		
		Diameter	Cross-section	Length
MIL-CuC-B	Half-round copper-coated carbon-graphite	3/8, 1/2, 5/8	Semi-circular	18
MIL-CuC-C	Round copper-coated carbon-graphite with machined lock-joint ends	5/16, 3/8 1/2 5/8 3/4, 1	Circular	12 14 15 17
MIL-CuC-D	Rectangular copper-coated carbon-graphite	---	5/32x3/8 3/16x5/8	12

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-576 - Boxes, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585 - Boxes, Shipping, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665 - Boxes: Paperboard, Metal Edged and Components.
PPP-B-676 - Boxes, Setup.

MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR

Code of Federal Regulations, Title 29
Part 1910 - Occupational Safety and Health Standards

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN WELDING SOCIETY (AWS)

A3.0 - AWS Definitions - Welding and Cutting.
Z49.1 - Safety in Welding and Cutting.

(Application for copies should be addressed to the American Welding Society, Inc., 550 NW LeJeune Road, P.O. Box 351040, Miami, FL 33125.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT

National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 1616 "P" Street, N.W., Washington, DC 20036.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

EW 1 - Electric Arc-Welding Apparatus.

(Application for copies should be addressed to National Electrical Manufacturers Association, 2101 "L" Street, NW, Washington, DC 20037.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Definitions. For the purpose of this specification, the welding terms and definitions contained in AWS A3.0 shall apply.

3.2 Dimensional tolerances.

3.2.1 Length. The length of the electrodes shall be not less than the nominal length.

3.2.2 Diameter. The diameter of the electrodes shall vary not more than plus or minus 5 percent from the nominal diameter.

3.2.3 Thickness and width. For the rectangular cross-section electrodes, type MIL-CuC-D, the dimensions "a" (thickness) and "b" (width) shall not vary more than plus or minus 5 percent from the nominal dimensions.

3.2.4 Straightness. The electrodes shall not deviate more than 1/2 percent of length as measured in 4.3.2.2.

3.2.5 Taper.

3.2.5.1 Types MIL-C and MIL-CuC-A electrodes other than 1/8 inch diameter shall have one end tapered approximately 30 degrees with the axis of the electrode. The diameter of the point shall be 1/16-inch minimum.

3.2.5.2 Type MIL-CuC-C electrode shall have both ends machined to a matching taper to provide a self-holding joint (electrode to electrode) with sufficiently close tolerance to permit use without stub loss in excess of the performance requirements specified in 3.5.3.

3.3 Copper coating. Coated electrodes shall have a copper coating 0.003 $\pm$ 0.0015 inch deposited on all surfaces except on the machined joint surfaces of type MIL-CuC-C and the tapered end of type MIL-CuC-A.

3.4 Resistivity. The resistivity of all types of electrodes shall be not more than 0.0010 ohm inch. The copper-coated electrodes shall have the copper removed for the test of 4.4.2.

3.5 Performance requirements.

3.5.1 Type MIL-C electrodes, performance tested in accordance with 4.4.3, shall not spall nor crack during use or subsequent cooling.

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3.5.2 Types MIL-CuC-A, MIL-CuC-B and MIL-CuC-D electrodes, performance tested in accordance with 4.4.3, shall:

(a) not spall during use or subsequent cooling,
(b) not have more than 15 percent of the remaining copper surface blistered,
(c) not have an average copper burn back in excess of 3-1/2 inches per electrode measured to the nearest 1/8-inch. The copper burn back is the distance from the tip of the burned electrode to the copper coating.

3.5.3 Type MIL-CuC-C electrodes, performance tested in accordance with 4.4.3, shall:

(a) not spall during use or subsequent cooling,
(b) not have a stub loss more than 1-1/2 rod diameters,
(c) not have more than 15 percent of the remaining copper surface blistered,
(d) not have an average copper burn back in excess of 3-1/2 inches per electrode measured to the nearest 1/8-inch. The cumulative copper burn back is the distance from the tip of the burned rod to the copper coating.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.

4.2.1 Inspection lot.

4.2.1.1 Containers. For the purpose of sampling and inspection of containers, all packaging and packing offered to one consignee for delivery at one time shall be considered a lot.

4.2.1.2 Electrodes. For the purpose of sampling and inspection, all electrodes of one type and size produced from one batch mix and offered for delivery at one time shall be considered a lot.

4.2.2 Sampling for examination of containers. A random sample of shipping containers filled with packages of electrodes shall be selected in accordance with MIL-STD-105 at inspection level S-1 and an acceptable quality level (AQL) of 2.5 percent to verify compliance with 4.3.1. The number of shipping containers shall be the lot size. Upon satisfactory completion of

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examination of filled shipping containers, a representative sample of the filled packages shall be selected in accordance with MIL-STD-105 at inspection level S-1 and an AQL of 2.5 percent. The sample shall be examined for conformance to 4.3.1.

4.2.3 Sampling for examination of electrodes. A random sample of electrodes shall be selected in accordance with MIL-STD-105 at inspection level S-1 and an AQL of 2.5 percent. The samples shall be examined for uniformity, dimensional tolerances, straightness and taper specified in 4.3.2 through 4.3.2.3.2.

4.2.4 Sampling for tests. Four electrodes shall be selected from the samples of 4.2.3 and subjected to the copper coating and resistivity tests specified in 4.4.1 and 4.4.2. Four additional electrodes shall be selected from the samples of 4.2.3 and subjected to the applicable test specified in 4.4.3.

4.3 Quality conformance inspection.

4.3.1 Examination of packaging and packing. Each of the containers, both outer shipping containers and unit packages, in a sample selected in accordance with 4.2.2 shall be examined for defects of construction, for unsatisfactory markings, and for compliance with the requirements specified in section 5.

4.3.2 Examination of electrodes (visual and dimensional). Each of the sample electrodes selected in accordance with 4.2.3 shall be examined as specified in 4.3.2.1 through 4.3.2.3.

4.3.2.1 Dimensional tolerances. The length and diameter of each of the electrodes selected in accordance with 4.2.3 shall be measured to determine conformance with 3.2.1, 3.2.2 and 3.2.3, respectively.

4.3.2.2 Straightness. The straightness (see 3.2.4) shall be verified by laying the electrode upon a surface plate or other smooth level surface and rolling it. During rotation, it shall be observed to determine the extent of gapping or separation between the two members. Any separation shall be measured with a thickness gage to determine conformance with 3.2.4.

4.3.2.3 Taper.

4.3.2.3.1 Types MIL-C and MIL-CuC-C electrodes shall be measured to determine taper angle and point diameter in conformance with 3.2.5.1.

4.3.2.3.2 Type MIL-CuC-C electrode taper shall be evaluated for compliance with 3.2.5.2.

4.4 Tests.

4.4.1 Copper coating. For coated electrodes, coating thickness shall be measured by stripping the copper coating at the center and both ends of the electrodes and measuring the foil peelings or by measuring the bare and covered cross sections and dividing by two. Extent of copper coating and thickness shall meet the requirements specified in 3.3.

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4.4.2 Resistivity. The resistivity of all types of electrodes shall be measured by the voltmeter-ammeter method as specified in 4.4.2.1 through 4.4.2.4 for conformance to 3.4.

4.4.2.1 Equipment. The following equipment shall be used:

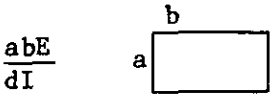
(a) Source of direct current (d.c.) providing a range of 6 volts (V) and 15 amperes (A) (for example, storage battery and variable resistance).
(b) Voltmeter. For the range of 5 to 200 millivolts (mV), a d.c. millivoltmeter having three scales of 0 to 20, 0 to 50, and 0 to 200 mV, or equivalent, shall be used. For the range of 0.4 to 20 mV, a d.c. millivoltmeter having four scales of 0 to 2, 0 to 5, 0 to 10 and 0 to 20 mV, or equivalent, shall be used.
(c) Ammeter. A d.c. ammeter having three scales of 0 to 1.5, 0 to 3.0, and 0 to 15, or equivalent, shall be used.
(d) Resilient contact surfaces such as silver plated copper gauze with a backing of resilient insulating material at least 0.25-inch thick.
(e) Sharp potential pointers arranged so as to contact a measured distance accurate to plus to minus 1/64-inch shall be used to obtain voltage measurement.

4.4.2.2 Test specimen. The copper coating shall be removed from the sample coated electrodes. Both ends of sample electrodes shall be cut through a full cross section perpendicular to the longitudinal axis.

4.4.2.3 Procedure. The test shall be conducted at 75 ± 15 degrees Fahrenheit (°F). A voltage shall be applied across the ends of the test electrode through resilient contact surfaces. The applied voltage shall be adjusted and meters selected so that the meter reading for the voltage drop across the 2-inch gage length and the current through the electrode will be at least 1/10 of the scale from either end of the range. Simultaneous voltage drop and current readings shall be made and recorded.

4.4.2.4 Calculation. The resistivity shall be calculated in accordance with the following formula:

Round	Half round	Rectangular
$K = \frac{3.1416R^2E}{dI}$	$K = \frac{3.1416R^2E}{2dI}$	$K = \frac{abE}{dI}$



Where:

K = resistivity (ohm inch).
R = radius of electrode (inches).
E = voltage drop between potential pointers in V.
d = gage length of potential pointer (inches).
I = current through specimen in A.
a = thickness.
b = width.

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4.4.3 Performance tests. Each of the four electrodes selected in accordance with 4.2.4 shall be subjected to the applicable test specified in 4.4.3.3, 4.4.3.4, or 4.4.3.5 for conformance to the requirements of 3.5.

4.4.3.1 Welding equipment. Welding machines used for supply power for testing electrodes shall be variable voltage d.c. motor-generator type or variable voltage d.c. welder conforming to NEMA EW 1. The power source shall be of sufficient rating to supply current demanded by the electrode type and size under test. Open circuit voltage shall not exceed 80 V.

4.4.3.2 Current levels. Current levels for the different electrodes shall be as specified in table II.

TABLE II. Current levels for performance test.

MIL-Type	Size (diameter) in inches 1/current in A									
	1/8	5/32	3/16	1/4	5/16	3/8	1/2	5/8	3/4	1
MIL-C	40	60	90	120	--	175	260	370	500	700
MIL-CuC-A	100	130	225	350	400	550	900	1100	1400	2000
MIL-CuC-B	--	--	--	--	--	225	350	400	--	--
MIL-CuC-C	--	--	--	--	400	550	900	1100	1400	2000

MIL-Type	Size (rectangular cross section (a x b) current in A (b / a))	
MIL-CuC-D	5/32 x 3/8	3/16 x 5/8
	200	400

1/ Current levels shall be maintained within plus or minus 10 percent of the listed values.

4.4.3.3 Performance of type MIL-C electrodes shall be determined by the following test:

(a) Place one end of electrode in holder. The holder shall be adapted with a nonconducting guide that protrudes to the mid length of the electrode and is alined with the test electrode.
(b) The test shall be conducted on the top horizontal surface of a plain carbon steel plate. The plate thickness shall be the manufacturer's choice based on the intended arc travel speed to prevent burn through.
(c) The electrode shall be tested at an angle of 45 ± 5 degrees with the plain carbon steel plate.
(d) The arc travel speed shall be the manufacturer's choice consistent with the plate thickness to prevent burn through.

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(e) The arc shall be struck and maintained until the nonconducting guide touches the plain carbon steel plate.
(f) Conformance to the performance requirements shall be determined during use and after the remaining portion of each of the four test electrodes has cooled.

4.4.3.4 Performance of type MIL-CuC-A, MIL-CuC-B, and MIL-CuC-D electrodes shall be determined by the following test:

(a) Place electrode in an air carbon-arc cutting holder with 8-3/4 to 9-1/4 inches of electrode protruding from the holder. The holder shall be adapted with a nonconducting guide that protrudes 5 inches from the torch and is alined with the test electrode.
(b) The test shall be conducted on the top horizontal surface of a plain carbon steel plate. The plate thickness shall be the manufacturer's choice based on the intended arc travel speed to prevent complete burn through (that is, to produce a gouging effect.)
(c) The test shall be conducted with 85 pounds per square inch (lb/in^2) air pressure plus or minus 5 lb/in^2. Air pressure to be measured at the hose connection not more than 8 feet from the torch with air at full flow.
(d) The test shall be conducted with the rod at 45 degree angle plus or minus 5 degrees with the plain carbon steel plate.
(e) The arc travel speed shall be the manufacturer's choice consistent with the plate thickness to prevent complete burn through.
(f) The arc shall be struck and maintained until the nonconducting guide touches the plain carbon steel plate.
(g) Conformance to the performance requirements shall be determined during use and after the remaining portion of each of the four test electrodes has cooled.

4.4.3.5 Performance of type MIL-CuC-C electrodes shall be determined by the following test:

(a) Cut off approximately a 4-inch length of electrode measured from tip of the female end and join the two pieces of the same rod. The joint shall be assembled with a slight twist (about 1/8-1/4 revolution), while holding the electrode pieces in alinement and firmly forcing them together.
(b) Place the joined electrode in an air carbon-arc cutting holder, with the cut end of the 4-inch length of electrode protruding 8-3/4 to 9-1/4 inches from the holder.
(c) The test shall be conducted on the top horizontal surface of a plain carbon steel plate. The plate thickness shall be the manufacturer's choice based on the intended arc travel speed to prevent complete burn through (that is, to produce a gouging effect).
(d) The test shall be conducted with 85 lb/in^2 air pressure plus or minus 5 lb/in^2. Air pressure shall be measured at the hose connection not more than 8 feet from the torch with air at full flow.

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(e) The test shall be conducted with the joined electrode at 45 degree angle plus or minus 5 degrees with the plain carbon steel plate.
(f) The arc travel speed shall be the manufacturer's choice consistent with the plate thickness to prevent complete burn through.
(g) The arc shall be struck on the cut end of the 4-inch length of electrode and kept in continuous use until the 4-inch length of electrode has been consumed.
(h) Conformance to the performance requirements shall be determined during use and after the remaining portion of each of the four joined test electrodes has cooled. Any portion of the 4-inch cut length electrode not consumed in the arc during the test due to separation at the joint shall be considered to be a stub loss in conformance with 3.5.3.

4.5 Retest. In case of failure of an electrode to conform with any specified requirement, one retest using electrodes selected from the same sampling or inspection lot, as those of the initial test, will be permitted.

4.6 Certificate of test. A certificate of quality conformance tests, containing actual test and inspection results, for each lot included in a particular shipment shall be furnished to the consignee (see 6.2.2). The minimum quality conformance test result data required shall be as listed in figure 1 and may be submitted on the form as shown.

4.7 Inspection of packaging. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3).

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2.1).

5.1.1 Level A. Electrodes shall be unit packaged in quantities of 50. Containers shall be as follows.

5.1.1.1 Unit containers. Unit containers shall conform to any one of the following specifications at the option of the contractor:

Specification	Container
PPP-B-566	Box, paperboard folding
PPP-B-636	Box, fiberboard
PPP-B-665	Box, paperboard, metal-stayed
PPP-B-676	Box, setup

The gross weight of folding and setup paperboard boxes shall not exceed 5 pounds; metal-stayed boxes 10 pounds; and fiberboard boxes 40 pounds.

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5.1.1.1.1 Cushioning. Use of loose fill materials such as sawdust, excelsior or similar materials are prohibited except, when enclosed in a sealed waterproof barrier.

5.1.1.2 Intermediate containers. Unit packages weighing 5 pounds or less shall be intermediate packaged in boxes conforming to PPP-B-636. The gross weight of the intermediate box shall not exceed 20 pounds. When specified (see 6.2.1), intermediate fiberboard boxes may be used as the shipping container for level B packing (see 5.2).

5.1.2 Level C. Electrodes shall be unit packaged in quantities as specified in 5.1.1. Packaging shall afford electrode protection against corrosion and damage during shipment from the supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2.1).

5.2.1 Level A. Electrodes, packaged as specified (see 6.2.1), shall be packed in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Container	Type or class
PPP-B-576	Wood-cleated veneer, paper overlaid	Overseas, class 2
PPP-B-585	Wood-wirebound	Class 3
PPP-B-601	Wood, cleated-plywood	Overseas type
PPP-B-621	Wood, nailed and lock-corner	Overseas, class 2

5.2.1.1 Closure caseliners and gross weight. Shipping containers shall be closed, strapped or reinforced in accordance with the applicable container specification or appendix thereto. Shipping containers shall have caseliners conforming to MIL-L-10547. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

5.2.2 Level B. Electrodes, packaged as specified (see 6.2.1), shall be packed in containers as specified in 5.2.1 except that the containers shall be of the domestic type or class as applicable. In addition, the following containers may be used at the option of the contractor:

Specification	Container	Type or class
PPP-B-591	Fiberboard, wood cleated	Weather-resistant
PPP-B-636	Fiberboard	Weather-resistant
PPP-B-640	Fiberboard, corrugated, triple wall	Weather-resistant

5.2.2.1 Closure and gross weight. Shipping containers shall be closed, strapped or reinforced in accordance with the applicable container specification. Intermediate fiberboard boxes (see 5.1.1.2), closed and reinforced as specified herein and used as shipping containers need not be overpacked. The gross weight of wood or wood-cleated containers shall not exceed 200 pounds; fiberboard containers shall not exceed the weight limitations of the applicable container specification.

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5.2.3 Level C. Electrodes, packaged as specified (see 6.2.1), shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules or Regulations or other carrier rules as applicable to the mode of transportation.

5.2.4 Palletized and containerized unit loads. When specified (see 6.2.1), shipping containers shall be palletized or containerized in accordance with MIL-STD-147.

5.3 Marking. In addition to any special marking (see 5.3.1 and 6.2.1), interior packages (unit and intermediate), shipping containers, palletized and containerized unit loads shall be marked for shipment in accordance with MIL-STD-129.

5.3.1 Interior packages. Shipment marking information shall be provided on interior packages in accordance with the manufacturer's commercial practice. In addition to the lot identification required by 4.2.1, the information shall include MIL type, size, specification number, manufacturer's or distributor's name, date of manufacture (month/year), and manufacturer's or distributor's brand or type designation. All packages, or the smallest integral unit within a shipping container, shall carry the following warning label, or equivalent prominently displayed in legible type on the package:

WARNING: Protect yourself and others. Read and understand this label.

FUMES AND GASES can be dangerous to your health. ARC RAYS can injure eyes and burn skin. ELECTRIC SHOCK can kill.

Read and understand the manufacturer's instructions and your employer's safety practices.
Keep your head out of the fumes.
Use enough general ventilation or exhaust at the arc or both to keep fumes and gases from your breathing zone, and the general area.
Wear correct eye, ear and body protection.
Do not touch live electrical parts.
See American National Standard Z49.1 "Safety in Welding and Cutting" published by the American Welding Society, Inc., 550 NW LeJeune Road, P.O. Box 351040, Miami, Florida 33125; OSHA Safety and Health Standards, 29 CFR 1910 are for sale by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

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6. NOTES

6.1 Intended use.

6.1.1 Type MIL-C electrodes are intended for use with the carbon-arc welding process.

6.1.2 Type MIL-CuC-A, MIL-CuC-B, and MIL-CuC-D electrodes are particularly suited to air carbon arc cutting and gouging operations.

6.1.3 Type MIL-CuC-C electrodes are particularly suited to air carbon-arc cutting and gouging operations with little or no stub loss.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2).
(c) Levels of preservation, packaging and packing required (see 5.1, 5.2, 5.2.1, 5.2.2 and 5.2.3).
(d) When intermediate containers are to be used as a shipping container (see 5.1.1.2).
(e) Unit loads when required (see 5.2.4).
(f) Special marking (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.6	Certificate of compliance	DI-E-2121	----

(Data item descriptions related to this specificationm and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the

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Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract, regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AR
Navy - SH
Air Force - 11

Review activities:
Navy - AS, MC
Air Force - 99
DLA - GS

User activities:
Army - AT
Navy - YD

Preparing activity:
Navy - SH
(Project 3439-0444)

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CERTIFICATION OF QUALITY CONFORMANCE TESTS

Manufacturer or _____ Customer's name _____

Distributor _____ Customer's Order No. _____

Address _____

_____

Specification: MIL-E-17777D

Type: MIL- _____

Lot No. _____

Size:

Round - Diameter (d) and length



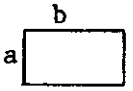
_____

Half Round - Diameter (d) and length



_____

Rectangular - a x b and length



_____

Straightness: _____ percent

Taper: _____ degrees

Copper coating thickness: _____

Resistivity: _____ ohm inch

Peformance test: _____ amperes

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements .

_____ _____

Signature of authorized company representative Date

FIGURE 1. Quality conformance test certification.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-17777D	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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